

Komal Jha

komaljha1504@gmail.com | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

Final-year Computer Science student at VIT Vellore targeting **AI Engineer** roles. Builds, evaluates and deploys **machine-learning** systems, from **PyTorch transfer learning** with **explainability (Grad-CAM)** to **scikit-learn** pipelines with rigorous **cross-validation**, backed by strong software engineering in **Python** and a frontend internship at **Tech Mahindra**.

Education

Vellore Institute of Technology, Vellore — <i>B.Tech in Computer Science (CGPA: 7.51)</i>	2023 – 2027
Swarajya Senior Secondary School, Alwar — <i>Class XII (77.8%)</i>	2023
Saint Francis School B.Deoghar, Deoghar — <i>Class X (95.4%)</i>	2021

Experience

Tech Mahindra, Navi Mumbai	May 2025 – July 2025
----------------------------	----------------------

Frontend Development Intern (7-week summer internship)

- **Contributed** to the frontend of a Corporate Banking Solutions application for a PSU bank.
- **Developed** and **debugged** UI components using **Angular**, **TypeScript**, **HTML** and **CSS**.
- **Resolved** styling issues across existing screens to improve UI consistency and responsiveness.
- **Collaborated** with mentors and team members to deliver enhancements aligned with business requirements.
- **Followed Agile** practices and collaborative workflows in an enterprise codebase.
- **Gained** exposure to corporate software workflows and collaborative development practices.

Projects

Dental X-ray Caries Classifier Code · Live demo	2026
---	------

Deep-learning image classifier with explainable heatmaps for panoramic dental X-rays

Tech Stack: Python, PyTorch, torchvision, OpenCV, NumPy, scikit-learn, Streamlit

- **Fine-tuned** ResNet-18 via **transfer learning** on 5,000 X-rays; reached **ROC-AUC 0.74** on 750 unseen images.
- **Implemented Grad-CAM** from scratch to explain every prediction and verify the model focuses on the teeth.
- **Preprocessed** images (224×224, ImageNet normalisation) with **augmentation** and a **stratified 70/15/15** split.
- **Trained** with **Adam**, **LR scheduling** and **early stopping**, freezing early layers to train 2× faster on CPU.
- **Evaluated** with accuracy, precision, recall, F1, confusion matrix and **bootstrap 95% CIs**.
- **Audited** the data, removed a **source confound**, and documented limitations honestly.
- **Tested** a 224×448 aspect-ratio input, disproved the hypothesis (AUC 0.69), and kept the stronger model.
- **Deployed** a live **Streamlit** demo returning class probabilities and a Grad-CAM overlay per upload.

Heart Disease Prediction Pipeline Code · Live demo	2026
--	------

End-to-end tabular ML pipeline on the UCI Heart Disease dataset with an interactive risk dashboard

Tech Stack: Python, scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Altair, Streamlit

- **Performed** EDA and **imputed** missing values, then **encoded** and **scaled** features inside a **leak-free pipeline**.
- **Engineered** 4 clinical features and tested each by **ablation** against the baseline.
- **Compared** 5 models incl. **XGBoost** under identical repeated **stratified CV** (5-fold × 3 repeats).
- **Tuned** the top 2 with **nested CV**; best model reached **ROC-AUC 0.92 ± 0.04** and **85.8% accuracy**.
- **Explained** predictions with **permutation importance** and per-patient factor attributions.
- **Reported** negative results honestly: tuning and engineered features gave no measurable gain.
- **Deployed** a live, interactive **Streamlit** app with a risk gauge and a full results dashboard.

Skills

- **Languages:** Python, Java, JavaScript, SQL, C++, C
- **ML & Deep Learning:** PyTorch, torchvision, scikit-learn, XGBoost, Transfer Learning, CNNs, Grad-CAM
- **Data & Evaluation:** Pandas, NumPy, Matplotlib, Cross-Validation, Hyperparameter Tuning, ROC-AUC, Feature Importance
- **Web & Backend:** Angular, React, Node.js, Express, REST APIs, MongoDB, PostgreSQL, Redis
- **Tools & Platforms:** Git, Streamlit, Gradio, Postman, Vercel, Render
- **Core Concepts:** Data Structures & Algorithms, OOP, System Design, Agile

Certifications

AWS Certified Cloud Practitioner — Amazon Web Services	2026
Data Structures & Algorithms — Udemy	2026